

Electrical properties and structure of lead-borate glass containing iron ions



N.A. Szreder, R.J. Barczyński*, J. Karczewski, M. Gazda

Department of Solid State Physics, Faculty of Applied Physics and Mathematics, Gdansk University of Technology (GUT), Narutowicza Street 11/12, Gdańsk 80-233, Poland

ARTICLE INFO

Article history:

Received 16 May 2013

Received in revised form 29 October 2013

Accepted 18 January 2014

Available online 13 February 2014

Keywords:

Iron-borate glass

Small-polaron hopping

Nanostructures

Impedance spectroscopy

Universal dielectric response

ABSTRACT

The ac and dc conductivity in iron-lead-borate glass samples was investigated in the frequency range from 1 mHz to 1 MHz and in the temperature range from 153 K to 423 K. The structure was investigated by the means of atomic force microscopy (AFM) and the crystalline phases (if present) were identified by the means of X-ray diffractometry. Two types of ac electrical behaviour were observed. The first group of glass samples which contain up to 31% mol Fe₂O₃ shows singular flattened semicircle in the impedance spectra. The second group of glass with over 25% mol Fe₂O₃ shows two distinguishable semicircles indicating two relaxation processes.

The results of structure investigation show that glass samples containing more than 2.5% mol Fe₂O₃ contain ball-shaped nanostructures. They have average size of an order of magnitude 100 nm. In XRD patterns for glass containing less than 19% mol Fe₂O₃ there are no diffraction peaks while in the patterns for samples with a larger amount of iron oxide some small peaks are present which indicate that some crystalline phases exist in these samples. In glass containing up to 19% iron oxide one process is responsible for conduction mechanism, small-polaron hopping. In the glass samples containing more of iron oxide, two dielectric relaxation processes were found. The second process may be caused by interface effect or conductivity process in the granules.

© 2014 Elsevier B.V. All rights reserved.

1. Introduction

The oxide glasses containing a large amount of transition metal oxides are electronic semiconductors which electrical conductivity is determined by the transition metal ions present in two different valence states. The structure and electrical properties of one group form this family – iron-lead-borate glasses were studied by Ardelean et al. [1–6]. They studied glass of several compositions with a different amount of Fe₂O₃ and the PbO/B₂O₃ ratio of 1:1 and 1:3. Their results for xFe₂O₃ (1 – x)(3B₂O₃–PbO) system suggested an amorphous nature of these glasses in the whole glass-formation region with the content of Fe₂O₃ up to 30% mol and 65% mol (for 1:1 and 1:3 PbO/B₂O₃ ratio respectively). On the basis of magnetic studies, they suggested that iron ions do not form clusters but are uniformly distributed in the glass matrix even at high Fe₂O₃ content [1,2]. They also examined electrical dc conductivity of xFe₂O₃ (1 – x)(3B₂O₃–PbO) glasses which for glasses containing over 15% mol Fe₂O₃ have two different activation energies one at the higher and second at the lower temperature regions. Only a single conduction mechanism was assumed, the small polaron hopping. As a reason for the two distinct activation energy values a charge transfer between iron ions in a similar environment at low temperatures and in a different environment at higher temperatures [5] was suggested.

A research made by Mendiratta [7] on the xFe₂O₃ (1 – x)(2B₂O₃–PbO) glass system suggested a presence of semiconducting α-Fe₂O₃ crystalline phase in the glasses containing 55% and 60% mol of Fe ions. His studies also showed that conductivity process in these glasses has two different activation energies. Suggestion was given that the conduction in these glasses is a mixture of hopping over iron sites and extended state conduction through the islands of α-Fe₂O₃ with defects. The lower activation energy is due to the conduction through the crystalline microregions distributed randomly in amorphous matrix, and the second higher activation energy is caused by the hopping over iron sites within the amorphous matrix.

Fahmy et al. [8] and Moon et al. [9] studied the structure of xFe₂O₃, (1 – x)(4B₂O₃–BaO) glass system. They found that for glasses containing over 3% mol of Fe₂O₃ the Fe³⁺ ions are not randomly distributed in glass matrix but they are incorporated into the clusters. A hypothesis was given that in glasses with the content of Fe₂O₃ lower than 10% mol there are only two types of clusters, the diad and the triad. They suggested that the affinity between the ions is strong enough to sufficiently reduce the isolated ion concentration [8,9].

All three cited models of structure of iron borate glasses are different and have divergent suggestions. The aim of the present study is to investigate the microstructure of these materials by the means of advanced microscopy and relate it to their electrical properties obtained by the means of impedance spectroscopy.

* Corresponding author.

E-mail address: jasiu@mif.pg.gda.pl (R.J. Barczyński).

2. Experimental

Glass samples of composition of $x\text{Fe}_2\text{O}_3-(100-x)(\text{B}_2\text{O}_3-2\text{PbO})$ ($2.5 < x < 37$, in mol%) were prepared by the conventional melt quenching technique. The melting was conducted in alumina crucibles at the temperature of 1523 K. The melts were poured on a preheated (573 K) brass plate and pressed by an another plate to obtain flat circular samples of 1–2 mm thickness and 20–30 mm in diameter. The quenched pellets were annealed at the temperature of 673 K for 3 h and cooled with the furnace.

For the electrical measurements gold electrodes were evaporated at the preheated samples. Impedance measurements were carried out in the frequency range from 1 mHz to 1 MHz with the ac voltage of 1 V rms in the temperature range from 153 K to 423 K with the Concept 40 broadband dielectric spectrometer. The structure has been studied by the X-ray diffraction method with the use of a Philips X'Pert Pro MPD system with the $\text{CuK}\alpha$ radiation and by atomic force microscopy with NT-MDT Ntegra AFM microscope.

3. Results

The impedance spectra for the most glass samples at the whole temperature range consist of one flattened and slightly distorted semicircle. Fig. 1 shows an example Nyquist plot for the glass sample $7\text{Fe}_2\text{O}_3-93(\text{B}_2\text{O}_3-2\text{PbO})$. Easily distinguishable semicircles of two relaxation processes are observed for glasses containing 31% of Fe_2O_3 at temperatures higher than 273 K and 37% of Fe_2O_3 at the whole temperature range (Fig. 2).

Fig. 3a presents the frequency dependence of the real part of conductivity measured at different temperatures for the $19\text{Fe}_2\text{O}_3-81(\text{B}_2\text{O}_3-2\text{PbO})$ glass sample. Similar dependences are observed for glasses containing 2.5% and 7% Fe_2O_3 . The plot consists of two regions: a dc plateau dominating in low frequency region and an ac component increasing with frequency in a linear fashion (power law behaviour). Fig. 3b displays the real part of conductivity versus frequency and temperature for $31\text{Fe}_2\text{O}_3-69(\text{B}_2\text{O}_3-2\text{PbO})$ glass. It may be seen that an additional relaxation process appears on the ac component. Similar behaviour is observed for glasses with 25% and 37% of Fe_2O_3 .

Fig. 4 shows atomic force microscopy images of the $x\text{Fe}_2\text{O}_3-(100-x)(\text{B}_2\text{O}_3-2\text{PbO})$ glass fractures for $x = 13\%$, 25% and 37%. The AFM micrograph for glass containing 2.5% Fe_2O_3 presents homogenous structure. As can be seen from Fig. 4 glasses with 13%, 25% and 37% mol Fe_2O_3 contain a large number of nanogranules. The rest of glass samples present similar structure as shown in Fig. 4. The size of ball-shaped nanogranules depends on the amount of iron oxide and decreases from about 400 nm to less than 100 nm with the increase in Fe_2O_3 content.

X-ray diffraction patterns of example glass samples are shown in Fig. 5. It may be noticed that glasses containing over 19% of Fe_2O_3 are not entirely amorphous. Their X-ray patterns (Fig. 5b, c and d) include not only amorphous halo but also some crystalline peaks. The peaks correspond to XRD spectra of Fe_3O_4 and Fe_2O_3 . The quantity and intensity of peaks increase with the increase in amount of iron oxide. The XRD pattern for glass containing 19% of Fe_2O_3 (Fig. 5a) shows no peaks and

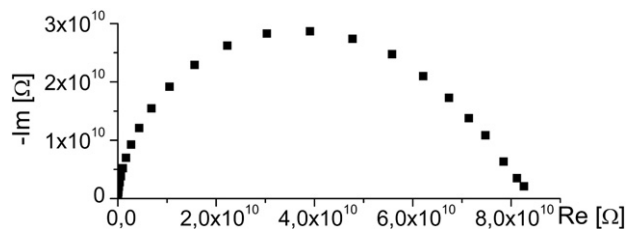


Fig. 1. The Nyquist plot for $7\text{Fe}_2\text{O}_3-93(\text{B}_2\text{O}_3-2\text{PbO})$ glass measured at 333 K.

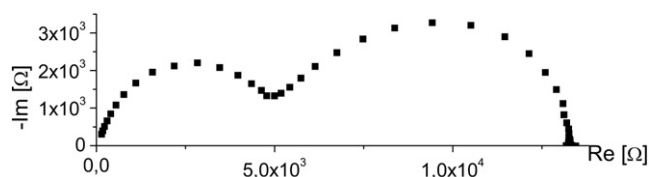


Fig. 2. The Nyquist plot for $37\text{Fe}_2\text{O}_3-63(\text{B}_2\text{O}_3-2\text{PbO})$ glass measured at 273 K.

suggests an amorphous structure. This situation is also observed in all glasses with less than 19% mol Fe_2O_3 .

4. Discussion

AFM micrographs (Fig. 4) and XRD spectra (Fig. 5) show that the structure of lead-iron-borate glasses depends on iron oxide content. These glasses may be amorphous and homogenous (less than 2.5% of Fe_2O_3), amorphous with spherical nanogranules (up to 25%) or may contain nanocrystallinities distributed in amorphous glass matrix (more than 25%). The Pb^{2+} ions may take part in the formation of the glass network because they are easily polarizable and exhibit covalent characteristics of bonding with oxygen. In lead-borate glasses the network may be formed from two different borate groups BO_3 and BO_4 , depending on concentration of lead in the glass matrix [10]. We assume that the amorphous glass matrix is formed by lead and borate oxides and iron ions are the structure modifiers. We also believe that the majority of iron ions in the glasses containing more than 2.5% Fe_2O_3 are located in nanostructures. In glasses with crystalline phase, the nanocrystallinities are formed by iron oxides: Fe_2O_3 and Fe_3O_4 . Our microphotographs of the samples of $x\text{Fe}_2\text{O}_3-(100-x)(\text{B}_2\text{O}_3-2\text{PbO})$ glass are in disagreement to Ardelean et al. proposal [1,2] which suggested that the structure is homogenous and iron ions are not divided into groups, regardless of the iron oxide content. They also does not support MacCrone et al. hypothesis [8,9] that iron ions are situated in clusters, but confirm the presence of nanostructures as Mendiratta studies suggested. It should be noted that the XRD pattern for glass containing 19% and less of Fe_2O_3 (Fig. 5a) shows no peaks despite the fact that their structure contains spherical granules.

Fig. 6 shows Arrhenius plot of dc conductivity versus $1000/T$ and determined activation energies for $x\text{Fe}_2\text{O}_3-(100-x)(\text{B}_2\text{O}_3-2\text{PbO})$ glasses where x equals 2.5%, 7%, 13%, 19%, 25%, 31% and 37%. It can be seen that glasses with over 13% mol Fe_2O_3 exhibit two distinctive values of the activation energy for high and low temperature regions. The activation energy decreases with the increase in iron oxide quantity and between 19% and 25% a steepest decrease is noticeable. This can be attributed to the development of the crystalline phase in the glass structure.

Table 1 shows the parameters of conductivity of all samples. The pre-exponential factor σ_0 fulfils the condition for non-adiabatic hopping [11,12] $\sigma_0 < (ne2R^2/kT)\nu_0$, where n is the estimated hopping centre concentration, R is the estimated distance between centres, and ν_0 is the phonon frequency. In this case the pre-exponential factor may be after Mott [13] expressed by $\sigma_0 = C(1-C)(e2vel/RkT)\exp(-2\alpha R)$, where C is the ratio of concentration of Fe^{2+} ions to total iron ion concentration ($C = [\text{Fe}^{2+}] / ([\text{Fe}^{2+}] + [\text{Fe}^{3+}])$), α is the wave function decay, vel is the electron frequency. On this basis α may be estimated, the obtained values are also given in Table 1.

In glass samples with a low iron oxide contents our electrical measurements are consistent with Ardelean et al. suggestion that the small polaron hopping is the only conduction mechanism. But two distinct activation energy values at different temperature ranges have an other reason than suggested by Ardelean. It is not the different environments of iron ions, but two different phases which are present in the glass-granules and the glass matrix. Mendiratta et al. suggested that conduction in these glasses is a mixture of hopping over iron sites and extended state conduction through islands of $\alpha\text{-Fe}_2\text{O}_3$ with defects. This model may be correct for glasses containing over 25% mol Fe_2O_3 when crystalline granules are observed.

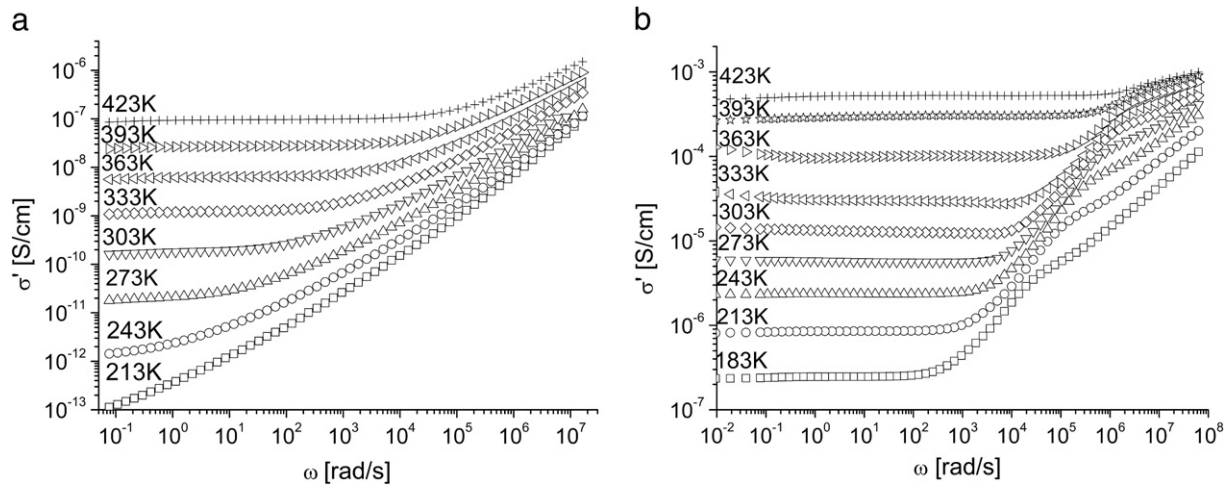


Fig. 3. Real part of conductivity versus frequency measured at different temperatures for (a) 19Fe₂O₃–81(B₂O₃–2PbO) and (b) 31Fe₂O₃–69(B₂O₃–2PbO) glass samples.

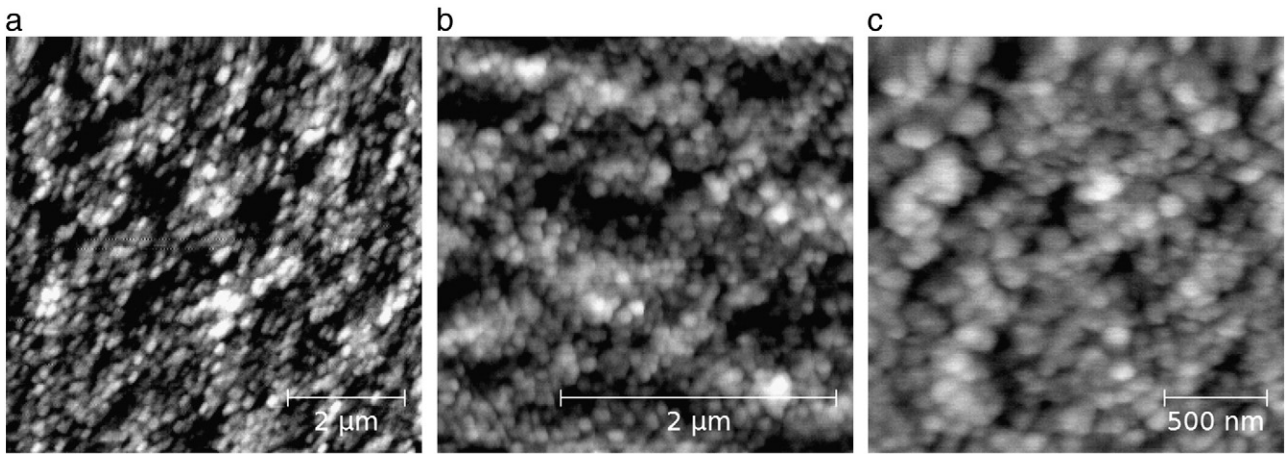


Fig. 4. Atomic force microscopy images of xFe₂O₃–(100 – x)(B₂O₃–2PbO) glass fractures in which the contents of Fe₂O₃ is respectively: (a) 13%, (b) 25% and (c) 37%.

5. Conclusion

The structure and electrical properties of lead-borate glasses containing iron ions were studied. It is found that the structure of glasses with more than 2.5% mol Fe₂O₃ is not homogenous and they contain a great number of nanogranules. Despite that no

peaks are visible in XRD pattern up to iron oxide contents of 19%. In glass containing more iron oxide the nanostructures are crystalline. Nanocrystallinities are randomly distributed in amorphous glass matrix and their size depends on the iron oxide amount.

Our electrical measurements are consistent with small polaron hopping mechanism. Two distinct activation energy values in glasses containing higher amount of iron ions at different temperature ranges are due to

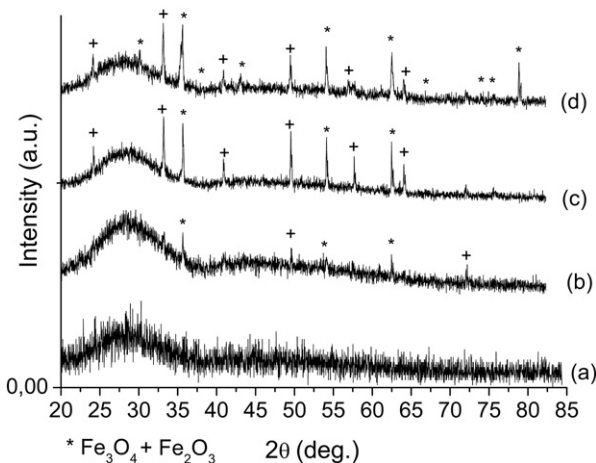


Fig. 5. XRD patterns for glass of composition: (a) 19Fe₂O₃–81(B₂O₃–2PbO), (b) 25Fe₂O₃–75(B₂O₃–2PbO), (c) 31Fe₂O₃–69(B₂O₃–2PbO) and (d) 37Fe₂O₃–63(B₂O₃–2PbO).

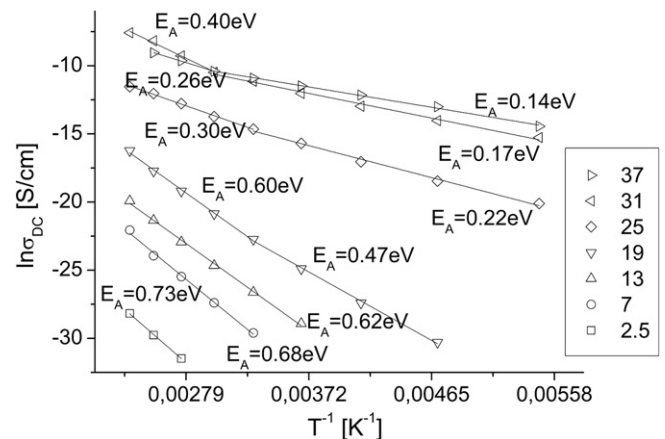


Fig. 6. Dc conductivity versus $1/T$ for xFe₂O₃–(100 – x)(B₂O₃–2PbO) glasses where x is 2.5%, 7%, 13%, 19%, 25%, 31% and 37%.

Table 1
Comparison of parameters of conductivity.

Composition [% mol]			σ_{Dc} (T = 393 K) [S/cm]		WDc [eV]	σ_0 [S/cm]	α (T = 393 K) [$\text{\AA}^{-1}$]		Structure	
Fe ₂ O ₃	B ₂ O ₃	PbO								
2.5	32.5	65	$1.18 \cdot 10^{-13}$		0.73	$2.55 \cdot 10^{-4}$	–		Amorphous and homogenous	
7	31	62	$4.02 \cdot 10^{-11}$		0.68	$2.80 \cdot 10^{-2}$	0.54		Amorphous with nanogranules	
13	29	58	$5.39 \cdot 10^{-10}$		0.62	$2.31 \cdot 10^{-2}$	0.69		Amorphous with nanogranules	
			Low T	High T	Low T	High T	Low T	High T		
19	27	54	$2.00 \cdot 10^{-8}$	0.47	0.60	$6.57 \cdot 10^{-3}$	$9.94 \cdot 10^{-1}$	0.91	0.60	Amorphous with nanogranules
25	25	50	$5.91 \cdot 10^{-6}$	0.22	0.30	$1.56 \cdot 10^{-3}$	$3.18 \cdot 10^{-2}$	1.14	0.93	Nanocrystallinities
31	23	46	$2.83 \cdot 10^{-4}$	0.17	0.40	$8.47 \cdot 10^{-3}$	$3.23 \cdot 10^{-1}$	1.11	0.48	Nanocrystallinities
37	21	42	$1.17 \cdot 10^{-4}$	0.14	0.26	$3.78 \cdot 10^{-3}$	$2.32 \cdot 10^{-1}$	1.28	0.94	Nanocrystallinities

two different phases present in the glass. A mixture of hopping over iron sites and extended state conduction through islands of $\alpha\text{-Fe}_2\text{O}_3$ with defects may be the charge transfer mechanism for glasses containing over 25% of Fe₂O₃.

References

- [1] I. Ardelean, E. Burzo, I. Pop, Solid State Commun. 23 (1977) 211–214.
- [2] I. Ardelean, Solid State Commun. 27 (1978) 697–703.
- [3] I. Ursu, E. Burzo, D. Ungur, I. Ardelean, Rev. Roum. Phys. 26 (1982) 801–813.
- [4] E. Burzo, D. Ungur, I. Ardelean, Solid State Commun. 50 (1984) 1083–1085.
- [5] E. Burzo, I. Ardelean, D. Ungur, Solid State Commun. 53 (1985) 283–285.
- [6] E. Burzo, D. Ungur, I. Ardelean, J. Phys. 44 (1983) 647–650.
- [7] S.K. Mendiratta, Phys. Stat. Sol. 93 (1986) 293.
- [8] M. Fahmy, M.J. Park, M. Tomazawa, R.K. MacCrone, Phys. Chem. Glas. 13 (1972) 21–26.
- [9] D.W. Moon, J.M. Aitken, R.K. MacCrone, G.S. Cieloszyk, Phys. Chem. Glasses 16 (1975) 91–102.
- [10] A.K. Jonscher, Dielectric Relaxation in Solids, Chelsea Dielectrics, London, 1983.
- [11] D. Emin, Electronic and Structural Properties of Amorphous Semiconductors, NY Academic Press, 1973. 261.
- [12] T. Holstein, Ann. Phys. 8 (325) (1959) 343.
- [13] N.F. Mott, E.A. Davis, Electron Processes in Non-Crystalline Materials, Oxford Clarendon Press, 1979.